Descriptive Statistics

Selected textbook reading

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OpenIntro Statistics, Fourth Edition (2019; updated 2022-10-21)
David Diez, Mine Çetinkaya-Rundel, Christopher D Barr

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Original PDF / printed pages 45-60 and 70-78. Sections 2.1.1-2.1.6 and 2.2.

Added this source notice; extracted selected pages without changing their content. Omitted advanced transformations, spatial maps, end-of-section exercises and the inference case study.

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2.1 Examining numerical data

In this section we will explore techniques for summarizing numerical variables. For example, consider the `loan_amount` variable from the `loan50` data set, which represents the loan size for all 50 loans in the data set. This variable is numerical since we can sensibly discuss the numerical difference of the size of two loans. On the other hand, area codes and zip codes are not numerical, but rather they are categorical variables.

Throughout this section and the next, we will apply these methods using the `loan50` and `county` data sets, which were introduced in Section 1.2. If you'd like to review the variables from either data set, see Figures 1.3 and 1.5.

2.1.1 Scatterplots for paired data

A **scatterplot** provides a case-by-case view of data for two numerical variables. In Figure 1.8 on page 18, a scatterplot was used to examine the homeownership rate against the fraction of housing units that were part of multi-unit properties (e.g. apartments) in the `county` data set. Another scatterplot is shown in Figure 2.1, comparing the total income of a borrower (`total_income`) and the amount they borrowed (`loan_amount`) for the `loan50` data set. In any scatterplot, each point represents a single case. Since there are 50 cases in `loan50`, there are 50 points in Figure 2.1.

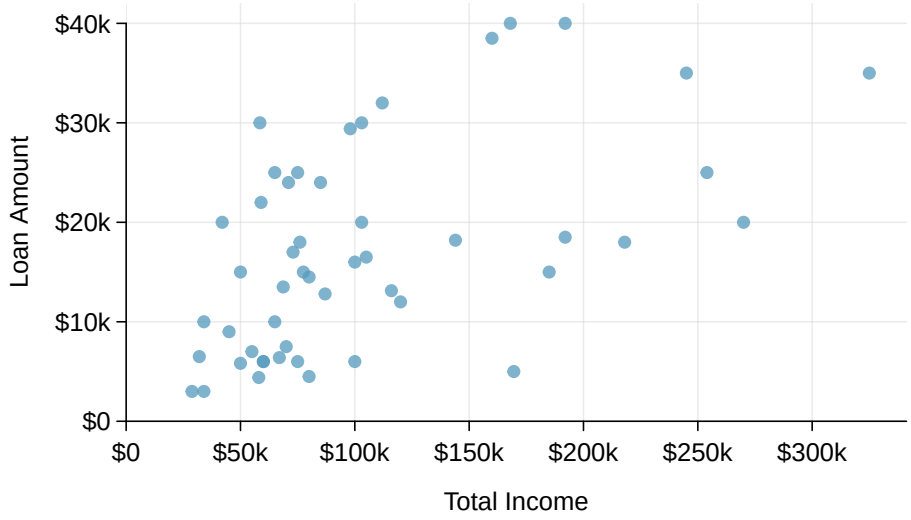


Figure 2.1: A scatterplot of `total_income` versus `loan_amount` for the `loan50` data set.

Looking at Figure 2.1, we see that there are many borrowers with an income below $100,000 on the left side of the graph, while there are a handful of borrowers with income above $250,000.

EXAMPLE 2.1 START

Example problem: Figure 2.2 shows a plot of median household income against the poverty rate for 3,142 counties. What can be said about the relationship between these variables?

Solution to the example: The relationship is evidently **nonlinear**, as highlighted by the dashed line. This is different from previous scatterplots we've seen, which show relationships that do not show much, if any, curvature in the trend.

EXAMPLE 2.1 HAS ENDED.

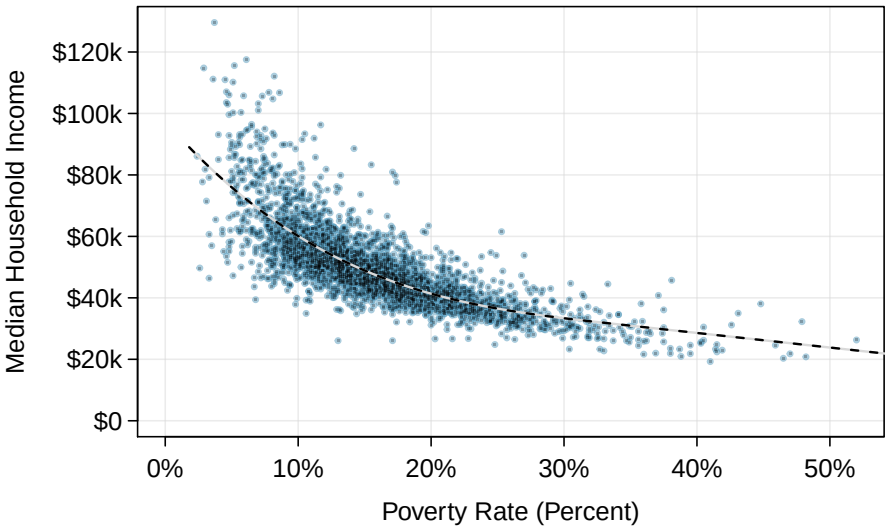


Figure 2.2: A scatterplot of the median household income against the poverty rate for the `county` data set. A statistical model has also been fit to the data and is shown as a dashed line.

GUIDED PRACTICE 2.2 START

What do scatterplots reveal about the data, and how are they useful?[1] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.2 HAS ENDED.

GUIDED PRACTICE 2.3 START

Describe two variables that would have a horseshoe-shaped association in a scatterplot ($\cap$ or $\frown$).[2] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.3 HAS ENDED.

2.1.2 Dot plots and the mean

Sometimes two variables are one too many: only one variable may be of interest. In these cases, a dot plot provides the most basic of displays. A **dot plot** is a one-variable scatterplot; an example using the interest rate of 50 loans is shown in Figure 2.3. A stacked version of this dot plot is shown in Figure 2.4.

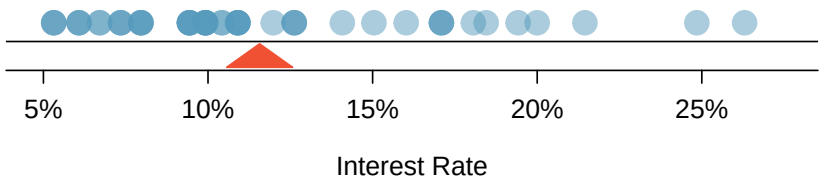


Figure 2.3: A dot plot of `interest_rate` for the `loan50` data set. The distribution's mean is shown as a red triangle.

[1] Answers may vary. Scatterplots are helpful in quickly spotting associations relating variables, whether those associations come in the form of simple trends or whether those relationships are more complex.

[2] Consider the case where your vertical axis represents something "good" and your horizontal axis represents something that is only good in moderation. Health and water consumption fit this description: we require some water to survive, but consume too much and it becomes toxic and can kill a person.

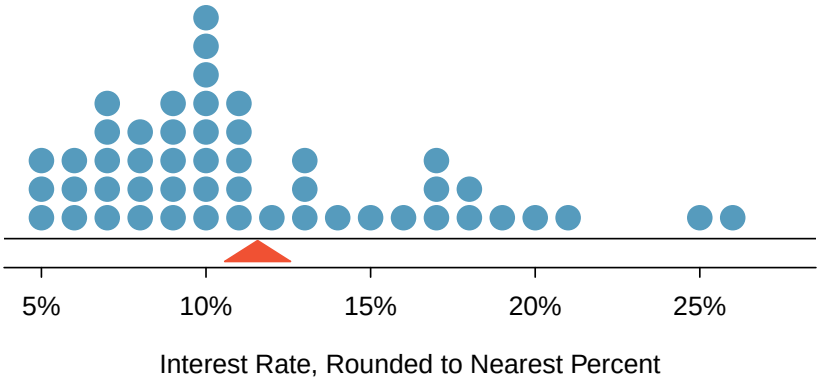


Figure 2.4: A stacked dot plot of `interest_rate` for the `loan50` data set. The rates have been rounded to the nearest percent in this plot, and the distribution's mean is shown as a red triangle.

The **mean**, often called the **average**, is a common way to measure the center of a **distribution** of data. To compute the mean interest rate, we add up all the interest rates and divide by the number of observations:

$$\bar{x} = \frac{10.90\% + 9.92\% + 26.30\% + \cdots + 6.08\%}{50} = 11.57\%$$

The sample mean is often labeled $\bar{x}$. The letter x is being used as a generic placeholder for the variable of interest, `interest_rate`, and the bar over the x communicates we're looking at the average interest rate, which for these 50 loans was 11.57%. It is useful to think of the mean as the balancing point of the distribution, and it's shown as a triangle in Figures 2.3 and 2.4.

MEAN

The sample mean can be computed as the sum of the observed values divided by the number of observations:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

where x_1, x_2, ..., x_n represent the n observed values.

GUIDED PRACTICE 2.4 START

Examine the equation for the mean. What does x_1 correspond to? And x_2? Can you infer a general meaning to what x_i might represent?[3] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.4 HAS ENDED.

GUIDED PRACTICE 2.5 START

What was n in this sample of loans?[4] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.5 HAS ENDED.

The `loan50` data set represents a sample from a larger population of loans made through Lending Club. We could compute a mean for this population in the same way as the sample mean. However, the population mean has a special label: μ. The symbol μ is the Greek letter *mu* and

[3] x_1 corresponds to the interest rate for the first loan in the sample (10.90%), x_2 to the second loan's interest rate (9.92%), and x_i corresponds to the interest rate for the ith loan in the data set. For example, if $i = 4$, then we're examining x_4, which refers to the fourth observation in the data set.

[4] The sample size was $n = 50$.

represents the average of all observations in the population. Sometimes a subscript, such as $_x$, is used to represent which variable the population mean refers to, e.g. μ_x. Often times it is too expensive to measure the population mean precisely, so we often estimate μ using the sample mean, $\bar{x}$.

EXAMPLE 2.6 START

Example problem: The average interest rate across all loans in the population can be estimated using the sample data. Based on the sample of 50 loans, what would be a reasonable estimate of μ_x, the mean interest rate for all loans in the full data set?

Solution to the example: The sample mean, 11.57%, provides a rough estimate of μ_x. While it's not perfect, this is our single best guess of the average interest rate of all the loans in the population under study.

In Chapter 5 and beyond, we will develop tools to characterize the accuracy of *point estimates* like the sample mean. As you might have guessed, point estimates based on larger samples tend to be more accurate than those based on smaller samples.

EXAMPLE 2.6 HAS ENDED.

EXAMPLE 2.7 START

Example problem: The mean is useful because it allows us to rescale or standardize a metric into something more easily interpretable and comparable. Provide 2 examples where the mean is useful for making comparisons.

Solution to the example:

1. We would like to understand if a new drug is more effective at treating asthma attacks than the standard drug. A trial of 1500 adults is set up, where 500 receive the new drug, and 1000 receive a standard drug in the control group:

	New drug	Standard drug
Number of patients	500	1000
Total asthma attacks	200	300

Comparing the raw counts of 200 to 300 asthma attacks would make it appear that the new drug is better, but this is an artifact of the imbalanced group sizes. Instead, we should look at the average number of asthma attacks per patient in each group:

$$\text{New drug: } 200/500 = 0.4 \qquad\qquad \text{Standard drug: } 300/1000 = 0.3$$

The standard drug has a lower average number of asthma attacks per patient than the average in the treatment group.

2. Emilio opened a food truck last year where he sells burritos, and his business has stabilized over the last 3 months. Over that 3 month period, he has made \$11,000 while working 625 hours. Emilio's average hourly earnings provides a useful statistic for evaluating whether his venture is, at least from a financial perspective, worth it:

$$\frac{\$11000}{625 \text{ hours}} = \$17.60 \text{ per hour}$$

By knowing his average hourly wage, Emilio now has put his earnings into a standard unit that is easier to compare with many other jobs that he might consider.

EXAMPLE 2.7 HAS ENDED.

EXAMPLE 2.8 START

Example problem: Suppose we want to compute the average income per person in the US. To do so, we might first think to take the mean of the per capita incomes across the 3,142 counties in the `county` data set. What would be a better approach?

Solution to the example: The `county` data set is special in that each county actually represents many individual people. If we were to simply average across the `income` variable, we would be treating counties with 5,000 and 5,000,000 residents equally in the calculations. Instead, we should compute the total income for each county, add up all the counties' totals, and then divide by the number of people in all the counties. If we completed these steps with the `county` data, we would find that the per capita income for the US is $30,861. Had we computed the *simple* mean of per capita income across counties, the result would have been just $26,093!

This example used what is called a **weighted mean**. For more information on this topic, check out the following online supplement regarding weighted means openintro.org/d?file=stat_wtd_mean.

EXAMPLE 2.8 HAS ENDED.

2.1.3 Histograms and shape

Dot plots show the exact value for each observation. This is useful for small data sets, but they can become hard to read with larger samples. Rather than showing the value of each observation, we prefer to think of the value as belonging to a *bin*. For example, in the `loan50` data set, we created a table of counts for the number of loans with interest rates between 5.0% and 7.5%, then the number of loans with rates between 7.5% and 10.0%, and so on. Observations that fall on the boundary of a bin (e.g. 10.00%) are allocated to the lower bin. This tabulation is shown in Figure 2.5. These binned counts are plotted as bars in Figure 2.6 into what is called a **histogram**, which resembles a more heavily binned version of the stacked dot plot shown in Figure 2.4.

Interest Rate	5.0% - 7.5%	7.5% - 10.0%	10.0% - 12.5%	12.5% - 15.0%	···	25.0% - 27.5%
Count	11	15	8	4	···	1

Figure 2.5: Counts for the binned `interest_rate` data.

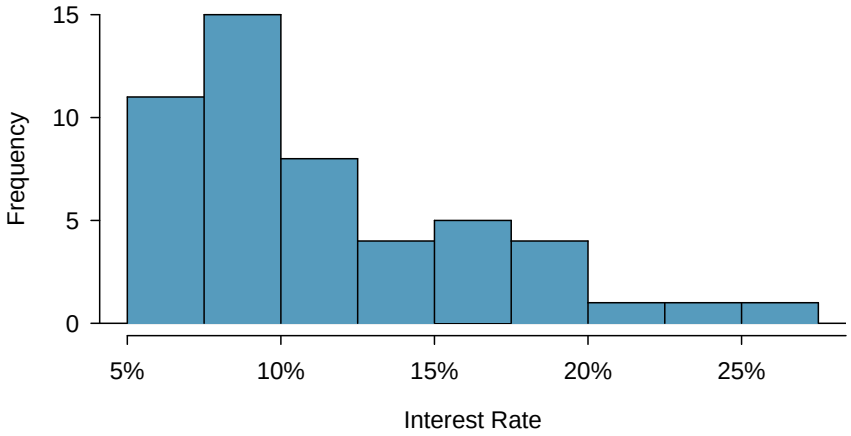


Figure 2.6: A histogram of `interest_rate`. This distribution is strongly skewed to the right.

Histograms provide a view of the **data density**. Higher bars represent where the data are relatively more common. For instance, there are many more loans with rates between 5% and 10% than loans with rates between 20% and 25% in the data set. The bars make it easy to see how the density of the data changes relative to the interest rate.

Histograms are especially convenient for understanding the shape of the data distribution. Figure 2.6 suggests that most loans have rates under 15%, while only a handful of loans have rates

above 20%. When data trail off to the right in this way and has a longer right tail, the shape is said to be **right skewed**.[5]

Data sets with the reverse characteristic – a long, thinner tail to the left – are said to be **left skewed**. We also say that such a distribution has a long left tail. Data sets that show roughly equal trailing off in both directions are called **symmetric**.

LONG TAILS TO IDENTIFY SKEW

When data trail off in one direction, the distribution has a **long tail**. If a distribution has a long left tail, it is left skewed. If a distribution has a long right tail, it is right skewed.

[5]Other ways to describe data that are right skewed: **skewed to the right**, **skewed to the high end**, or **skewed to the positive end**.

GUIDED PRACTICE 2.9 START

Take a look at the dot plots in Figures 2.3 and 2.4. Can you see the skew in the data? Is it easier to see the skew in this histogram or the dot plots?[6] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.9 HAS ENDED.

GUIDED PRACTICE 2.10 START

Besides the mean (since it was labeled), what can you see in the dot plots that you cannot see in the histogram?[7] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.10 HAS ENDED.

In addition to looking at whether a distribution is skewed or symmetric, histograms can be used to identify modes. A **mode** is represented by a prominent peak in the distribution. There is only one prominent peak in the histogram of `loan_amount`.

A definition of *mode* sometimes taught in math classes is the value with the most occurrences in the data set. However, for many real-world data sets, it is common to have *no* observations with the same value in a data set, making this definition impractical in data analysis.

Figure 2.7 shows histograms that have one, two, or three prominent peaks. Such distributions are called **unimodal**, **bimodal**, and **multimodal**, respectively. Any distribution with more than 2 prominent peaks is called multimodal. Notice that there was one prominent peak in the unimodal distribution with a second less prominent peak that was not counted since it only differs from its neighboring bins by a few observations.

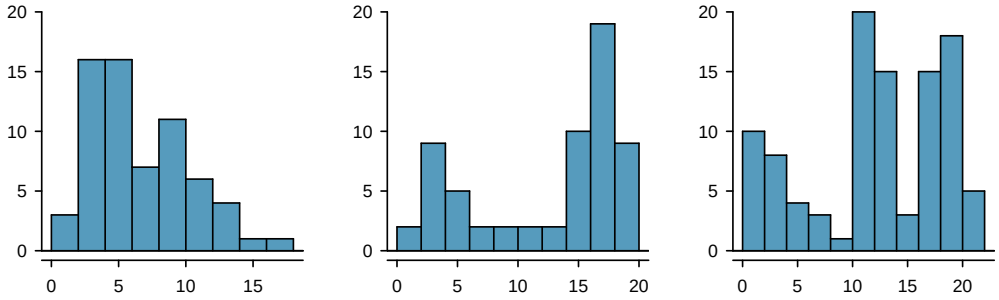


Figure 2.7: Counting only prominent peaks, the distributions are (left to right) unimodal, bimodal, and multimodal. Note that we've said the left plot is unimodal intentionally. This is because we are counting *prominent* peaks, not just any peak.

EXAMPLE 2.11 START

Example problem: Figure 2.6 reveals only one prominent mode in the interest rate. Is the distribution unimodal, bimodal, or multimodal?

Solution to the example: Unimodal. Remember that *uni* stands for 1 (think *uni*cycles). Similarly, *bi* stands for 2 (think *bi*cycles). We're hoping a *multicycle* will be invented to complete this analogy.

EXAMPLE 2.11 HAS ENDED.

GUIDED PRACTICE 2.12 START

Height measurements of young students and adult teachers at a K-3 elementary school were taken. How many modes would you expect in this height data set?[8] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.12 HAS ENDED.

[6] The skew is visible in all three plots, though the flat dot plot is the least useful. The stacked dot plot and histogram are helpful visualizations for identifying skew.

[7] The interest rates for individual loans.

Looking for modes isn't about finding a clear and correct answer about the number of modes in a distribution, which is why *prominent* is not rigorously defined in this book. The most important part of this examination is to better understand your data.

[8]There might be two height groups visible in the data set: one of the students and one of the adults. That is, the data are probably bimodal.

2.1.4 Variance and standard deviation

The mean was introduced as a method to describe the center of a data set, and variability in the data is also important. Here, we introduce two measures of variability: the variance and the standard deviation. Both of these are very useful in data analysis, even though their formulas are a bit tedious to calculate by hand. The standard deviation is the easier of the two to comprehend, and it roughly describes how far away the typical observation is from the mean.

We call the distance of an observation from its mean its **deviation**. Below are the deviations for the 1st, 2nd, 3rd, and 50th observations in the `interest_rate` variable:

$$\begin{aligned} x_1 - \bar{x} &= 10.90 - 11.57 = -0.67 \\ x_2 - \bar{x} &= 9.92 - 11.57 = -1.65 \\ x_3 - \bar{x} &= 26.30 - 11.57 = 14.73 \\ &\vdots \\ x_{50} - \bar{x} &= 6.08 - 11.57 = -5.49 \end{aligned}$$

If we square these deviations and then take an average, the result is equal to the sample **variance**, denoted by s^2:

$$\begin{aligned} s^2 &= \frac{(-0.67)^2 + (-1.65)^2 + (14.73)^2 + \cdots + (-5.49)^2}{50 - 1} \\ &= \frac{0.45 + 2.72 + 216.97 + \cdots + 30.14}{49} \\ &= 25.52 \end{aligned}$$

We divide by $n - 1$, rather than dividing by n, when computing a sample's variance; there's some mathematical nuance here, but the end result is that doing this makes this statistic slightly more reliable and useful.

Notice that squaring the deviations does two things. First, it makes large values relatively much larger, seen by comparing $(-0.67)^2$, $(-1.65)^2$, $(14.73)^2$, and $(-5.49)^2$. Second, it gets rid of any negative signs.

The **standard deviation** is defined as the square root of the variance:

$$s = \sqrt{25.52} = 5.05$$

While often omitted, a subscript of $_x$ may be added to the variance and standard deviation, i.e. s_x^2 and s_x, if it is useful as a reminder that these are the variance and standard deviation of the observations represented by x_1, x_2, ..., x_n.

VARIANCE AND STANDARD DEVIATION

The variance is the average squared distance from the mean. The standard deviation is the square root of the variance. The standard deviation is useful when considering how far the data are distributed from the mean.

The standard deviation represents the typical deviation of observations from the mean. Usually about 70% of the data will be within one standard deviation of the mean and about 95% will be within two standard deviations. However, as seen in Figures 2.8 and 2.9, these percentages are not strict rules.

Like the mean, the population values for variance and standard deviation have special symbols: σ^2 for the variance and σ for the standard deviation. The symbol σ is the Greek letter *sigma*.

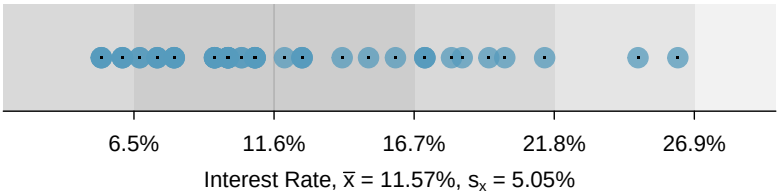


Figure 2.8: For the `interest_rate` variable, 34 of the 50 loans (68%) had interest rates within 1 standard deviation of the mean, and 48 of the 50 loans (96%) had rates within 2 standard deviations. Usually about 70% of the data are within 1 standard deviation of the mean and 95% within 2 standard deviations, though this is far from a hard rule.

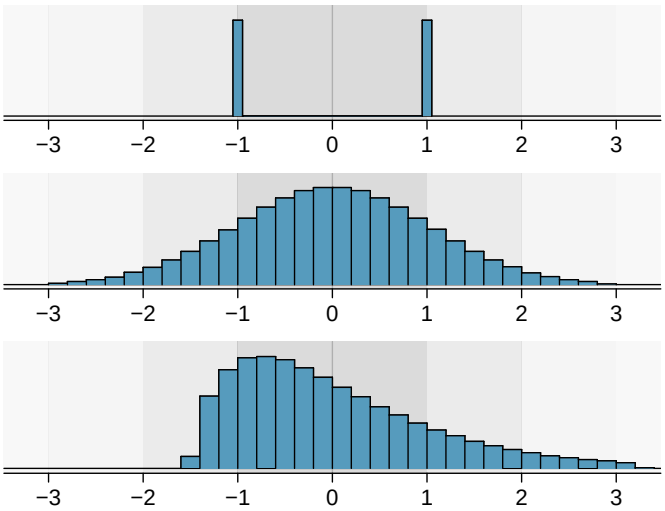


Figure 2.9: Three very different population distributions with the same mean $\mu = 0$ and standard deviation $\sigma = 1$.

GUIDED PRACTICE 2.13 START

On page 50, the concept of shape of a distribution was introduced. A good description of the shape of a distribution should include modality and whether the distribution is symmetric or skewed to one side. Using Figure 2.9 as an example, explain why such a description is important.[9] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.13 HAS ENDED.

EXAMPLE 2.14 START

Example problem: Describe the distribution of the `interest_rate` variable using the histogram in Figure 2.6. The description should incorporate the center, variability, and shape of the distribution, and it should also be placed in context. Also note any especially unusual cases.

Solution to the example: The distribution of interest rates is unimodal and skewed to the high end. Many of the rates fall near the mean at 11.57%, and most fall within one standard deviation (5.05%) of the mean. There are a few exceptionally large interest rates in the sample that are above 20%.

EXAMPLE 2.14 HAS ENDED.

In practice, the variance and standard deviation are sometimes used as a means to an end, where

[9] Figure 2.9 shows three distributions that look quite different, but all have the same mean, variance, and standard deviation. Using modality, we can distinguish between the first plot (bimodal) and the last two (unimodal). Using skewness, we can distinguish between the last plot (right skewed) and the first two. While a picture, like a histogram, tells a more complete story, we can use modality and shape (symmetry/skew) to characterize basic information about a distribution.

the "end" is being able to accurately estimate the uncertainty associated with a sample statistic. For example, in Chapter 5 the standard deviation is used in calculations that help us understand how much a sample mean varies from one sample to the next.

2.1.5 Box plots, quartiles, and the median

A **box plot** summarizes a data set using five statistics while also plotting unusual observations. Figure 2.10 provides a vertical dot plot alongside a box plot of the `interest_rate` variable from the `loan50` data set.

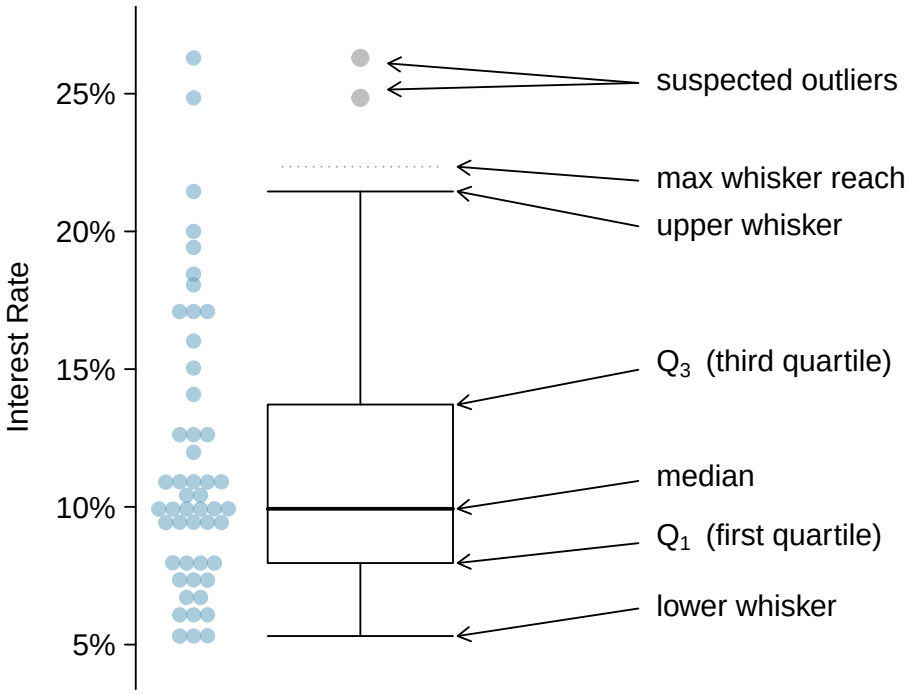


Figure 2.10: A vertical dot plot, where points have been horizontally stacked, next to a labeled box plot for the interest rates of the 50 loans.

The first step in building a box plot is drawing a dark line denoting the **median**, which splits the data in half. Figure 2.10 shows 50% of the data falling below the median and other 50% falling above the median. There are 50 loans in the data set (an even number) so the data are perfectly split into two groups of 25. We take the median in this case to be the average of the two observations closest to the 50th percentile, which happen to be the same value in this data set: $(9.93\% + 9.93\%)/2 = 9.93\%$. When there are an odd number of observations, there will be exactly one observation that splits the data into two halves, and in such a case that observation is the median (no average needed).

MEDIAN: THE NUMBER IN THE MIDDLE

If the data are ordered from smallest to largest, the **median** is the observation right in the middle. If there are an even number of observations, there will be two values in the middle, and the median is taken as their average.

The second step in building a box plot is drawing a rectangle to represent the middle 50% of the data. The total length of the box, shown vertically in Figure 2.10, is called the **interquartile range** (IQR, for short). It, like the standard deviation, is a measure of variability in data. The more variable the data, the larger the standard deviation and IQR tend to be. The two boundaries of the box are called the **first quartile** (the 25th percentile, i.e. 25% of the data fall below this value) and the **third quartile** (the 75th percentile), and these are often labeled Q_1 and Q_3, respectively.

INTERQUARTILE RANGE (IQR)

The IQR is the length of the box in a box plot. It is computed as

$$IQR = Q_3 - Q_1$$

where Q_1 and Q_3 are the 25th and 75th percentiles.

GUIDED PRACTICE 2.15 START

What percent of the data fall between Q_1 and the median? What percent is between the median and Q_3?[10] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.15 HAS ENDED.

Extending out from the box, the **whiskers** attempt to capture the data outside of the box. However, their reach is never allowed to be more than $1.5 \times IQR$. They capture everything within this reach. In Figure 2.10, the upper whisker does not extend to the last two points, which is beyond $Q_3 + 1.5 \times IQR$, and so it extends only to the last point below this limit. The lower whisker stops at the lowest value, 5.31%, since there is no additional data to reach; the lower whisker's limit is not shown in the figure because the plot does not extend down to $Q_1 - 1.5 \times IQR$. In a sense, the box is like the body of the box plot and the whiskers are like its arms trying to reach the rest of the data.

Any observation lying beyond the whiskers is labeled with a dot. The purpose of labeling these points – instead of extending the whiskers to the minimum and maximum observed values – is to help identify any observations that appear to be unusually distant from the rest of the data. Unusually distant observations are called **outliers**. In this case, it would be reasonable to classify the interest rates of 24.85% and 26.30% as outliers since they are numerically distant from most of the data.

OUTLIERS ARE EXTREME

An **outlier** is an observation that appears extreme relative to the rest of the data.

Examining data for outliers serves many useful purposes, including

1. Identifying strong skew in the distribution.
2. Identifying possible data collection or data entry errors.
3. Providing insight into interesting properties of the data.

GUIDED PRACTICE 2.16 START

Using Figure 2.10, estimate the following values for `interest_rate` in the `loan50` data set:
(a) Q_1, (b) Q_3, and (c) IQR.[11] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.16 HAS ENDED.

[10] Since Q_1 and Q_3 capture the middle 50% of the data and the median splits the data in the middle, 25% of the data fall between Q_1 and the median, and another 25% falls between the median and Q_3.

[11] These visual estimates will vary a little from one person to the next: $Q_1 = 8\%$, $Q_3 = 14\%$, IQR $= Q_3 - Q_1 = 6\%$. (The true values: $Q_1 = 7.96\%$, $Q_3 = 13.72\%$, IQR $= 5.76\%$.)

2.1.6 Robust statistics

How are the sample statistics of the `interest_rate` data set affected by the observation, 26.3%? What would have happened if this loan had instead been only 15%? What would happen to these summary statistics if the observation at 26.3% had been even larger, say 35%? These scenarios are plotted alongside the original data in Figure 2.11, and sample statistics are computed under each scenario in Figure 2.12.

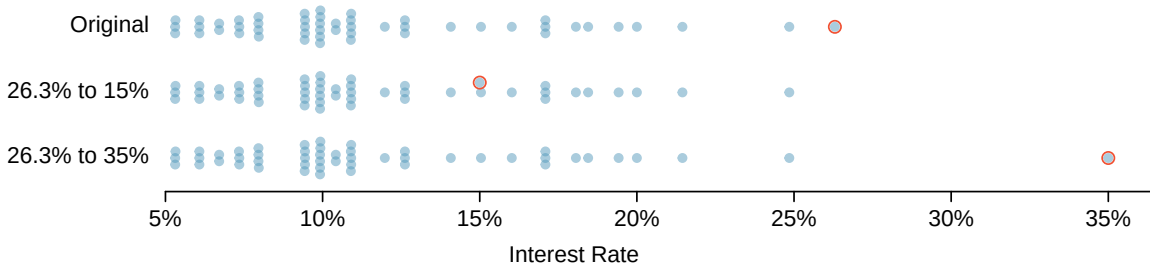


Figure 2.11: Dot plots of the original interest rate data and two modified data sets.

	robust		not robust	
scenario	median	IQR	$\bar{x}$	s
original `interest_rate` data	9.93%	5.76%	11.57%	5.05%
move 26.3% $\rightarrow$ 15%	9.93%	5.76%	11.34%	4.61%
move 26.3% $\rightarrow$ 35%	9.93%	5.76%	11.74%	5.68%

Figure 2.12: A comparison of how the median, IQR, mean ($\bar{x}$), and standard deviation (s) change had an extreme observations from the `interest_rate` variable been different.

GUIDED PRACTICE 2.17 START

(a) Which is more affected by extreme observations, the mean or median? Figure 2.12 may be helpful. (b) Is the standard deviation or IQR more affected by extreme observations?[12] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.17 HAS ENDED.

The median and IQR are called **robust statistics** because extreme observations have little effect on their values: moving the most extreme value generally has little influence on these statistics. On the other hand, the mean and standard deviation are more heavily influenced by changes in extreme observations, which can be important in some situations.

EXAMPLE 2.18 START

Example problem: The median and IQR did not change under the three scenarios in Figure 2.12. Why might this be the case?

Solution to the example: The median and IQR are only sensitive to numbers near Q_1, the median, and Q_3. Since values in these regions are stable in the three data sets, the median and IQR estimates are also stable.

EXAMPLE 2.18 HAS ENDED.

[12] (a) Mean is affected more. (b) Standard deviation is affected more. Complete explanations are provided in the material following Guided Practice 2.17.

GUIDED PRACTICE 2.19 START

The distribution of loan amounts in the `loan50` data set is right skewed, with a few large loans lingering out into the right tail. If you were wanting to understand the typical loan size, should you be more interested in the mean or median?[13] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.19 HAS ENDED.

[13] Answers will vary! If we're looking to simply understand what a typical individual loan looks like, the median is probably more useful. However, if the goal is to understand something that scales well, such as the total amount of money we might need to have on hand if we were to offer 1,000 loans, then the mean would be more useful.

2.2 Considering categorical data

In this section, we will introduce tables and other basic tools for categorical data that are used throughout this book. The `loan50` data set represents a sample from a larger loan data set called `loans`. This larger data set contains information on 10,000 loans made through Lending Club. We will examine the relationship between `homeownership`, which for the `loans` data can take a value of `rent`, `mortgage` (owns but has a mortgage), or `own`, and `app_type`, which indicates whether the loan application was made with a partner or whether it was an individual application.

2.2.1 Contingency tables and bar plots

Figure 2.17 summarizes two variables: `app_type` and `homeownership`. A table that summarizes data for two categorical variables in this way is called a **contingency table**. Each value in the table represents the number of times a particular combination of variable outcomes occurred. For example, the value 3496 corresponds to the number of loans in the data set where the borrower rents their home and the application type was by an individual. Row and column totals are also included. The **row totals** provide the total counts across each row (e.g. $3496 + 3839 + 1170 = 8505$), and **column totals** are total counts down each column. We can also create a table that shows only the overall percentages or proportions for each combination of categories, or we can create a table for a single variable, such as the one shown in Figure 2.18 for the `homeownership` variable.

		homeownership			
		rent	mortgage	own	Total
app_type	individual	3496	3839	1170	8505
	joint	362	950	183	1495
	Total	3858	4789	1353	10000

Figure 2.17: A contingency table for `app_type` and `homeownership`.

homeownership	Count
rent	3858
mortgage	4789
own	1353
Total	10000

Figure 2.18: A table summarizing the frequencies of each value for the `homeownership` variable.

A bar plot is a common way to display a single categorical variable. The left panel of Figure 2.19 shows a **bar plot** for the `homeownership` variable. In the right panel, the counts are converted into proportions, showing the proportion of observations that are in each level (e.g. $3858/10000 = 0.3858$ for `rent`).

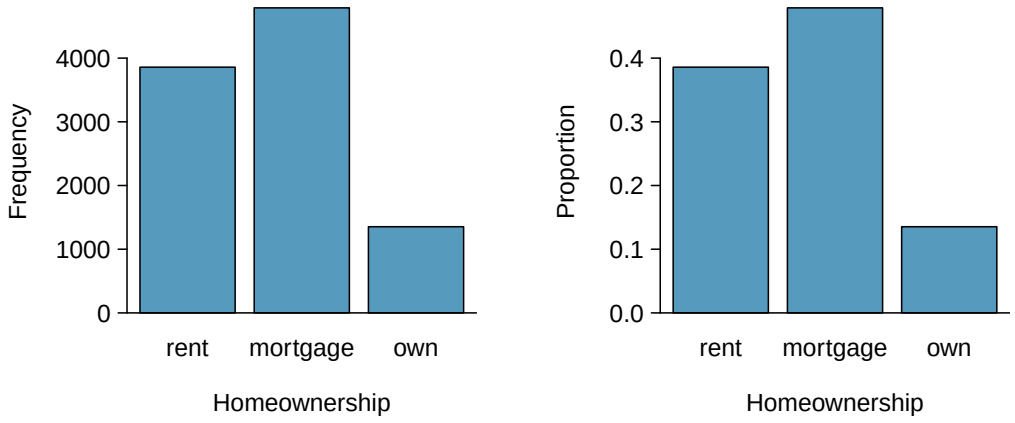


Figure 2.19: Two bar plots of `number`. The left panel shows the counts, and the right panel shows the proportions in each group.

2.2.2 Row and column proportions

Sometimes it is useful to understand the fractional breakdown of one variable in another, and we can modify our contingency table to provide such a view. Figure 2.20 shows the **row proportions** for Figure 2.17, which are computed as the counts divided by their row totals. The value 3496 at the intersection of `individual` and `rent` is replaced by 3496/8505 = 0.411, i.e. 3496 divided by its row total, 8505. So what does 0.411 represent? It corresponds to the proportion of individual applicants who rent.

	rent	mortgage	own	Total
individual	0.411	0.451	0.138	1.000
joint	0.242	0.635	0.122	1.000
Total	0.386	0.479	0.135	1.000

Figure 2.20: A contingency table with row proportions for the `app_type` and `homeownership` variables. The row total is off by 0.001 for the `joint` row due to a rounding error.

A contingency table of the column proportions is computed in a similar way, where each **column proportion** is computed as the count divided by the corresponding column total. Figure 2.21 shows such a table, and here the value 0.906 indicates that 90.6% of renters applied as individuals for the loan. This rate is higher compared to loans from people with mortgages (80.2%) or who own their home (86.5%). Because these rates vary between the three levels of `homeownership` (`rent`, `mortgage`, `own`), this provides evidence that the `app_type` and `homeownership` variables are associated.

	rent	mortgage	own	Total
individual	0.906	0.802	0.865	0.851
joint	0.094	0.198	0.135	0.150
Total	1.000	1.000	1.000	1.000

Figure 2.21: A contingency table with column proportions for the `app_type` and `homeownership` variables. The total for the last column is off by 0.001 due to a rounding error.

We could also have checked for an association between `app_type` and `homeownership` in Figure 2.20 using row proportions. When comparing these row proportions, we would look down columns to see if the fraction of loans where the borrower rents, has a mortgage, or owns varied across the `individual` to `joint` application types.

GUIDED PRACTICE 2.23 START

(a) What does 0.451 represent in Figure 2.20?
(b) What does 0.802 represent in Figure 2.21?[18] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.23 HAS ENDED.

GUIDED PRACTICE 2.24 START

(a) What does 0.122 at the intersection of `joint` and `own` represent in Figure 2.20?
(b) What does 0.135 represent in the Figure 2.21?[19] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.24 HAS ENDED.

EXAMPLE 2.25 START

Example problem: Data scientists use statistics to filter spam from incoming email messages. By noting specific characteristics of an email, a data scientist may be able to classify some emails as spam or not spam with high accuracy. One such characteristic is whether the email contains no numbers, small numbers, or big numbers. Another characteristic is the email format, which indicates whether or not an email has any HTML content, such as bolded text. We'll focus on email format and spam status using the `email` data set, and these variables are summarized in a contingency table in Figure 2.22. Which would be more helpful to someone hoping to classify email as spam or regular email for this table: row or column proportions?

Solution to the example: A data scientist would be interested in how the proportion of spam changes within each email format. This corresponds to column proportions: the proportion of spam in plain text emails and the proportion of spam in HTML emails.

If we generate the column proportions, we can see that a higher fraction of plain text emails are spam ($209/1195 = 17.5\%$) than compared to HTML emails ($158/2726 = 5.8\%$). This information on its own is insufficient to classify an email as spam or not spam, as over 80% of plain text emails are not spam. Yet, when we carefully combine this information with many other characteristics, we stand a reasonable chance of being able to classify some emails as spam or not spam with confidence.

EXAMPLE 2.25 HAS ENDED.

	text	HTML	Total
spam	209	158	367
not spam	986	2568	3554
Total	1195	2726	3921

Figure 2.22: A contingency table for `spam` and `format`.

Example 2.25 points out that row and column proportions are not equivalent. Before settling on one form for a table, it is important to consider each to ensure that the most useful table is constructed. However, sometimes it simply isn't clear which, if either, is more useful.

[18] (a) 0.451 represents the proportion of individual applicants who have a mortgage. (b) 0.802 represents the fraction of applicants with mortgages who applied as individuals.
[19] (a) 0.122 represents the fraction of joint borrowers who own their home. (b) 0.135 represents the home-owning borrowers who had a joint application for the loan.

EXAMPLE 2.26 START

Example problem: Look back to Tables 2.20 and 2.21. Are there any obvious scenarios where one might be more useful than the other?

Solution to the example: None that we thought were obvious! What is distinct about `app_type` and `homeownership` vs the email example is that these two variables don't have a clear explanatory-response variable relationship that we might hypothesize (see Section 1.2.4 for these terms). Usually it is most useful to "condition" on the explanatory variable. For instance, in the email example, the email format was seen as a possible explanatory variable of whether the message was spam, so we would find it more interesting to compute the relative frequencies (proportions) for each email format.

EXAMPLE 2.26 HAS ENDED.

2.2.3 Using a bar plot with two variables

Contingency tables using row or column proportions are especially useful for examining how two categorical variables are related. Stacked bar plots provide a way to visualize the information in these tables.

A **stacked bar plot** is a graphical display of contingency table information. For example, a stacked bar plot representing Figure 2.21 is shown in Figure 2.23(a), where we have first created a bar plot using the `homeownership` variable and then divided each group by the levels of `app_type`.

One related visualization to the stacked bar plot is the **side-by-side bar plot**, where an example is shown in Figure 2.23(b).

For the last type of bar plot we introduce, the column proportions for the `app_type` and `homeownership` contingency table have been translated into a standardized stacked bar plot in Figure 2.23(c). This type of visualization is helpful in understanding the fraction of individual or joint loan applications for borrowers in each level of `homeownership`. Additionally, since the proportions of `joint` and `individual` vary across the groups, we can conclude that the two variables are associated.

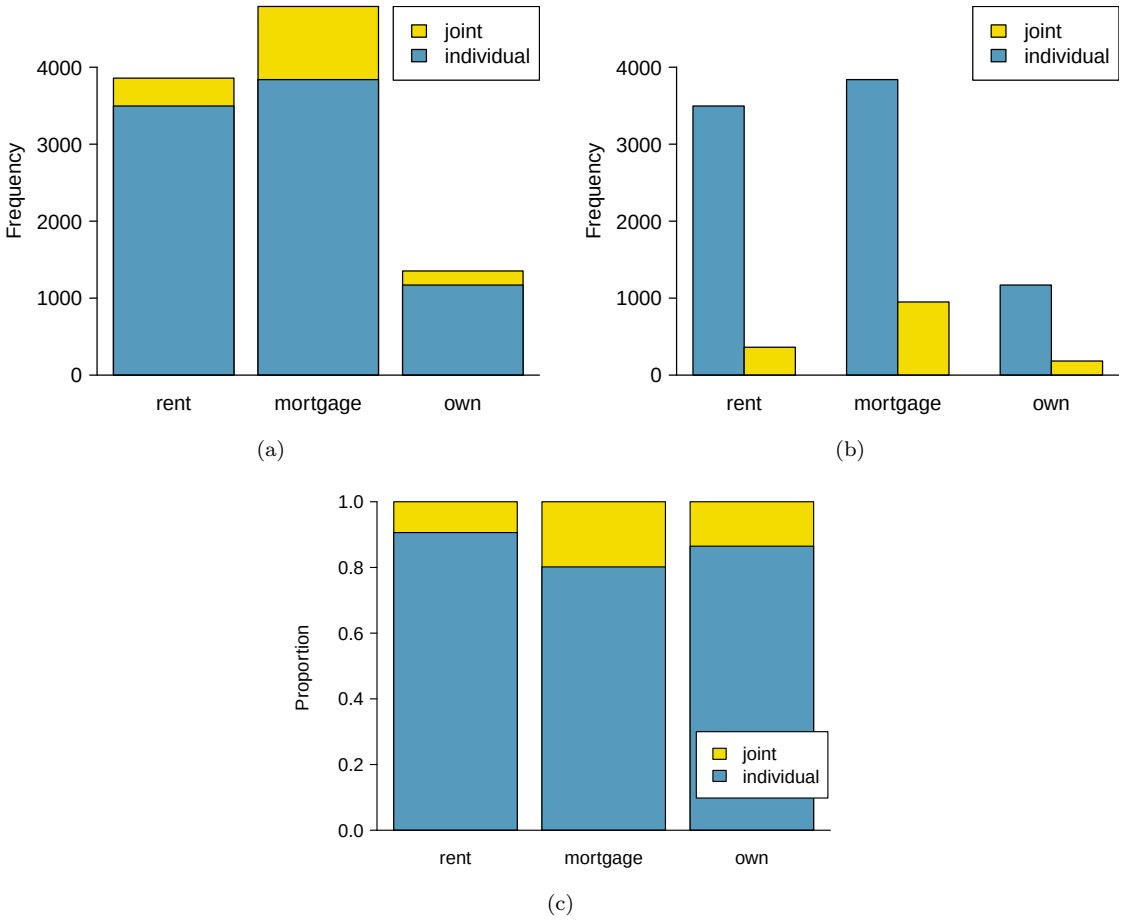


Figure 2.23: (a) Stacked bar plot for `homeownership`, where the counts have been further broken down by `app_type`. (b) Side-by-side bar plot for `homeownership` and `app_type`. (c) Standardized version of the stacked bar plot.

EXAMPLE 2.27 START

Example problem: Examine the three bar plots in Figure 2.23. When is the stacked, side-by-side, or standardized stacked bar plot the most useful?

Solution to the example: The stacked bar plot is most useful when it's reasonable to assign one variable as the explanatory variable and the other variable as the response, since we are effectively grouping by one variable first and then breaking it down by the others.

Side-by-side bar plots are more agnostic in their display about which variable, if any, represents the explanatory and which the response variable. It is also easy to discern the number of cases in the six different group combinations. However, one downside is that it tends to require more horizontal space; the narrowness of Figure 2.23(b) makes the plot feel a bit cramped. Additionally, when two groups are of very different sizes, as we see in the `own` group relative to either of the other two groups, it is difficult to discern if there is an association between the variables.

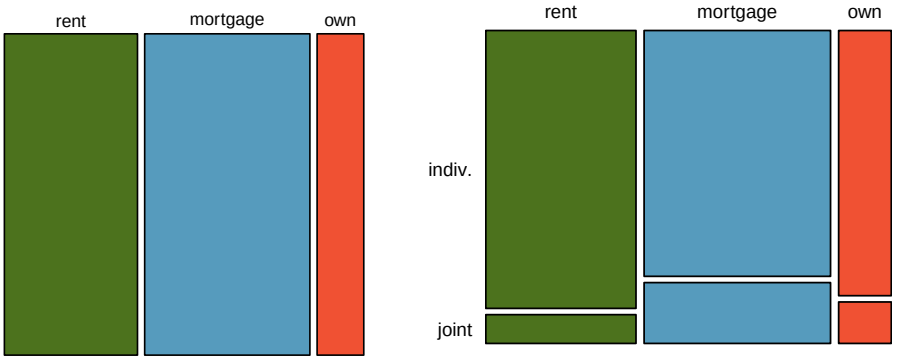
The standardized stacked bar plot is helpful if the primary variable in the stacked bar plot is relatively imbalanced, e.g. the `own` category has only a third of the observations in the `mortgage` category, making the simple stacked bar plot less useful for checking for an association. The major downside of the standardized version is that we lose all sense of how many cases each of the bars represents.

EXAMPLE 2.27 HAS ENDED.

2.2.4 Mosaic plots

A **mosaic plot** is a visualization technique suitable for contingency tables that resembles a standardized stacked bar plot with the benefit that we still see the relative group sizes of the primary variable as well.

To get started in creating our first mosaic plot, we'll break a square into columns for each category of the `homeownership` variable, with the result shown in Figure 2.24(a). Each column represents a level of `homeownership`, and the column widths correspond to the proportion of loans in each of those categories. For instance, there are fewer loans where the borrower is an owner than where the borrower has a mortgage. In general, mosaic plots use box *areas* to represent the number of cases in each category.



(a) (b)

Figure 2.24: (a) The one-variable mosaic plot for `homeownership`. (b) Two-variable mosaic plot for both `homeownership` and `app_type`.

To create a completed mosaic plot, the single-variable mosaic plot is further divided into pieces in Figure 2.24(b) using the `app_type` variable. Each column is split proportional to the number of loans from individual and joint borrowers. For example, the second column represents loans where the borrower has a mortgage, and it was divided into individual loans (upper) and joint loans (lower). As another example, the bottom segment of the third column represents loans where the borrower owns their home and applied jointly, while the upper segment of this column represents borrowers who are homeowners and filed individually. We can again use this plot to see that the

`homeownership` and `app_type` variables are associated, since some columns are divided in different vertical locations than others, which was the same technique used for checking an association in the standardized stacked bar plot.

In Figure 2.24, we chose to first split by the homeowner status of the borrower. However, we could have instead first split by the application type, as in Figure 2.25. Like with the bar plots, it's common to use the explanatory variable to represent the first split in a mosaic plot, and then for the response to break up each level of the explanatory variable, if these labels are reasonable to attach to the variables under consideration.

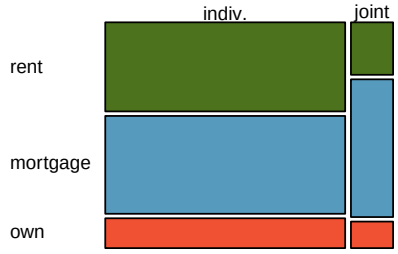


Figure 2.25: Mosaic plot where loans are grouped by the `homeownership` variable after they've been divided into the `individual` and `joint` application types.

2.2.5 The only pie chart you will see in this book

A **pie chart** is shown in Figure 2.26 alongside a bar plot representing the same information. Pie charts can be useful for giving a high-level overview to show how a set of cases break down. However, it is also difficult to decipher details in a pie chart. For example, it takes a couple seconds longer to recognize that there are more loans where the borrower has a mortgage than rent when looking at the pie chart, while this detail is very obvious in the bar plot. While pie charts can be useful, we prefer bar plots for their ease in comparing groups.

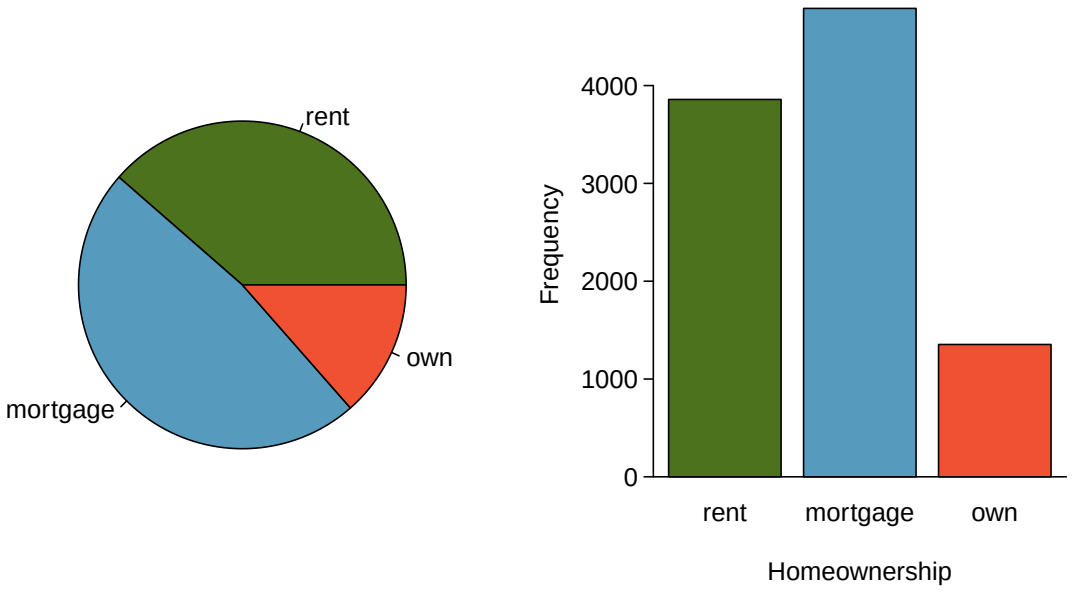


Figure 2.26: A pie chart and bar plot of `homeownership`.

2.2.6 Comparing numerical data across groups

Some of the more interesting investigations can be considered by examining numerical data across groups. The methods required here aren't really new: all that's required is to make a numerical plot for each group in the same graph. Here two convenient methods are introduced: side-by-side box plots and hollow histograms.

We will take a look again at the `county` data set and compare the median household income for counties that gained population from 2010 to 2017 versus counties that had no gain. While we might like to make a causal connection here, remember that these are observational data and so such an interpretation would be, at best, half-baked.

There were 1,454 counties where the population increased from 2010 to 2017, and there were 1,672 counties with no gain (all but one were a loss). A random sample of 100 counties from the first group and 50 from the second group are shown in Figure 2.27 to give a better sense of some of the raw median income data.

Median Income for 150 Counties, in $1000s

Population Gain						No Population Gain		
38.2	43.6	42.2	61.5	51.1	45.7	48.3	60.3	50.7
44.6	51.8	40.7	48.1	56.4	41.9	39.3	40.4	40.3
40.6	63.3	52.1	60.3	49.8	51.7	57	47.2	45.9
51.1	34.1	45.5	52.8	49.1	51	42.3	41.5	46.1
80.8	46.3	82.2	43.6	39.7	49.4	44.9	51.7	46.4
75.2	40.6	46.3	62.4	44.1	51.3	29.1	51.8	50.5
51.9	34.7	54	42.9	52.2	45.1	27	30.9	34.9
61	51.4	56.5	62	46	46.4	40.7	51.8	61.1
53.8	57.6	69.2	48.4	40.5	48.6	43.4	34.7	45.7
53.1	54.6	55	46.4	39.9	56.7	33.1	21	37
63	49.1	57.2	44.1	50	38.9	52	31.9	45.7
46.6	46.5	38.9	50.9	56	34.6	56.3	38.7	45.7
74.2	63	49.6	53.7	77.5	60	56.2	43	21.7
63.2	47.6	55.9	39.1	57.8	42.6	44.5	34.5	48.9
50.4	49	45.6	39	38.8	37.1	50.9	42.1	43.2
57.2	44.7	71.7	35.3	100.2		35.4	41.3	33.6
42.6	55.5	38.6	52.7	63		43.4	56.5	

Figure 2.27: In this table, median household income (in $1000s) from a random sample of 100 counties that had population gains are shown on the left. Median incomes from a random sample of 50 counties that had no population gain are shown on the right.

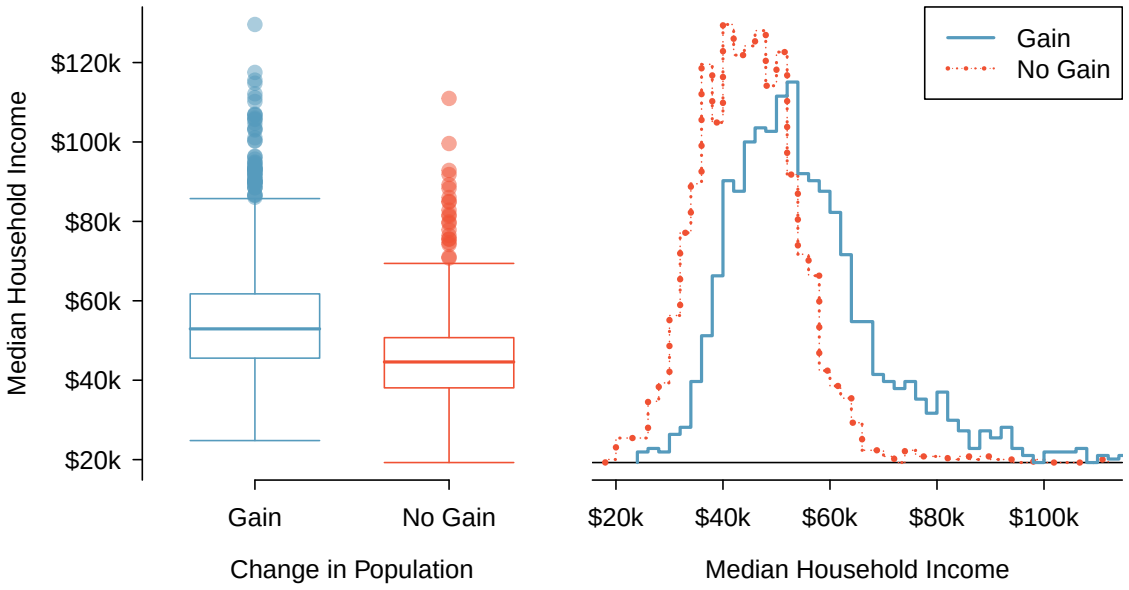


Figure 2.28: Side-by-side box plot (left panel) and hollow histograms (right panel) for `med_hh_income`, where the counties are split by whether there was a population gain or loss.

The **side-by-side box plot** is a traditional tool for comparing across groups. An example is shown in the left panel of Figure 2.28, where there are two box plots, one for each group, placed into one plotting window and drawn on the same scale.

Another useful plotting method uses **hollow histograms** to compare numerical data across groups. These are just the outlines of histograms of each group put on the same plot, as shown in the right panel of Figure 2.28.

GUIDED PRACTICE 2.28 START

Use the plots in Figure 2.28 to compare the incomes for counties across the two groups. What do you notice about the approximate center of each group? What do you notice about the variability between groups? Is the shape relatively consistent between groups? How many *prominent* modes are there for each group?[20] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.28 HAS ENDED.

GUIDED PRACTICE 2.29 START

What components of each plot in Figure 2.28 do you find most useful?[21] Go to the preceding footnote link for the Guided Practice solution.
GUIDED PRACTICE 2.29 HAS ENDED.

[20] Answers may vary a little. The counties with population gains tend to have higher income (median of about $45,000) versus counties without a gain (median of about $40,000). The variability is also slightly larger for the population gain group. This is evident in the IQR, which is about 50% bigger in the *gain* group. Both distributions show slight to moderate right skew and are unimodal. The box plots indicate there are many observations far above the median in each group, though we should anticipate that many observations will fall beyond the whiskers when examining any data set that contain more than a couple hundred data points.

[21] Answers will vary. The side-by-side box plots are especially useful for comparing centers and spreads, while the hollow histograms are more useful for seeing distribution shape, skew, and potential anomalies.